Reddit Crypto Discussion Sentiment Analysis

Team Data Envision-Airs

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https://github.com/uic-cs418/group-project-data-envision-airs

Introduction

This project explores whether public sentiment on Reddit influences cryptocurrency market performance. We're analyzing posts from eight crypto-related subreddits and financial data for five major cryptocurrencies (BTC, ETH, DOGE, SOL, XRP) from 03/20/2024 to 09/20/2024. Using sentiment analysis tools like VADER, we classify each post as positive, negative, or neutral, and then examine how those trends align with price changes. Our goal is to identify any patterns that might help investors better understand how online discussions impact the crypto market.

Data

Reddit Post Data

We obtained the Reddit post data from the Reddit API: https://www.reddit.com/dev/api/, scraping posts from r/Bitcoin, r/ethereum, r/dogecoin, r/solana, and r/XRP. The data was collected using praw Python Library, and using a python script located in `reddit.py` into files in reddit_posts folder pulling the top posts (sorted by highest "upvotes") from the past year for each subreddit.

Structure: The dataset has three columns: "*PostDate*": The date and time when the post was created, formatted as YYYY-MM-DD HH:MM:SS. "*PostTitle: Content*": A combined string of the post's title and content, separated by a colon. "*Upvotes*": The number of upvotes the post received.

Granularity Each row in the dataset represents a single Reddit post from a specific subreddit. The data includes details about the post's creation date, title and content (combined), and upvotes.

Scope The posts are gathered as the top posts from the last year from each subreddit. Because subreddit popularity varies, the number of posts scraped is slightly different for each:

Subreddit	Number of Posts
Bitcoin	1000
ethereum	866
dogecoin	999
solana	998
XRP	557

This gives a total of 4420 posts collected

Temporality and Faithfulness The data is directly fetched from Reddit using the official Reddit API through the praw library, ensuring high accuracy and reliability. The data reflects the actual posts and their metadata as they exist on Reddit at the time of collection from the "Top Posts" from 03/20/2024 to 09/20/2024.

Data CLeaning and Transformation

Combining Columns: The title and content of each post were combined into a single column ("PostTitle: Content") for simplicity.

Date Formatting: The post creation date was converted from a Unix timestamp to a human-readable format (YYYY-MM-DD HH:MM:SS).

File Organization: Separate CSV files were created for each subreddit to keep the data organized.

Missing or empty post content (selftext) was replaced with "N/A". This ensures that every row in the dataset has a value for the "PostTitle: Content" column, even if the post did not have any content. (Some posts are just a title with no content, in the same way a Piazza post could just be a title without content)

Cryptocurrency Data

CoinBase

We obtained the Coinbase crypto historical data from the Coinbase API: https://docs.cdp.coinbase.com/exchange/docs/welcome. We cleaned the data from the Coinbase API using a python script located in `pulling_coins.py` and storing it in `crypto_data` folder. From the API, We pulled the historical data with one pull per day at the 00:00:00 mark from 03/20/2024 to 09/20/2024.

Structure: The dataset has four columns: "*timestamp, low, high, open, close*": The date is formatted as YYYY-MM-DD. "*Low, high, and close*": are the prices that occured during that day with the closing being the final price before it was finalized for the day.

Granularity Each row in the dataset represents a single day of the coin historical data. The data includes details about the coins's up and downs of the day.

Scope The crypto data is gathered by date from 03/20/2024 to 09/20/2024 with 186 lines in the csv (one for headers).

Temporality and Faithfulness The data is directly fetched from CoinBase using the official CoinBase API through their public library, ensuring high accuracy and reliability. The data reflects the actual posts and their metadata as they exist on CoinBase, spanning six months of 2024.

Data CLeaning and Transformation

Date Formatting: The post creation date was converted from a Unix timestamp to a human-readable format (YYYY-MM-DD).

File Organization: Separate CSV files were created for each Coin to keep the data organized.

Missing or empty post content: There was none as it is historical financial data.

Machine Learning with VADER

The first ML Analysis we performed on our data was a sentiment analysis. To do this. We used VADER sentiment analysis tool to add tags of "postive", "negative", or "neutral" to each reddit post in each subreddit (see `ML_baseline_comparison/sentiment_analysis.py` for the script to generate the files located in `/reddit_posts_with_sentiment`).

Since this is unlabeled data (there are no "correct" labels to compare our sentiment analysis against), we performed a qualitative analysis to produce a baseline comparison to evaluate the accuracy of our sentiment analysis. To do this, we randomly selected 20 posts from each subreddit and read the title and post and assigned my own sentiment tag to them and compared whether the VADER sentiment analysis accuractely predicted the tone of the post.

Below you can see the number of positive/negative/neutral posts per subreddit.

Name, Total, Positive, Negative, Neutral Bitcoin, 1000, 420, 401, 179
ethereum, 999, 484, 410, 105
dogecoin, 866, 553, 176, 137
solana, 998, 596, 243, 159
XRP, 557, 343, 129, 85

As mentioned previously, to perform a baseline analysis, we randomly selected some posts to manually tag and compared them with the actual tags (located in `ML_baseline_comaprison/combined_samples.csv`). This reported an accuracy of **63%**. Though this may seem low, we are not entirely concerned about that due to the fact that positive/negative/neutral are fairly subjective. Our model correctly classifies (according to our judgement) more than half the time and we are satisfied with that accuracy.

Moving forward, the VADER labeled data seemed accurate enough to continue our project. In addition, the fact that there are varying amounts of positive posts means that there are likely periods of time where the overall sentiment of the subreddit changes, just as a cryptocurrency would. We will use this labeled data to compare change in public outlook vs actual cryptocurrency value changes.

Hypothesis Testing

In `percent_dif.py`, with the labeled reddit data, we calculated the percent change in the percent of positive posts per day. We also calculated the percent change in the cryptocurrency value per day. With those two values, we calculated the absolute difference. That absolute difference is what we used to perform a statistical analysis on each cryptocurrency. (results from the python script are location in `/percent_difs`)

In [17]:
```
# statisitcal analysis
import pandas as pd

# read the CSV file
df = pd.read_csv('./statistical_analysis/summary_stats.csv')

df
```

Out[17]:

	File	Min	Q1	Mean	Median	Mode	Standard Deviation	Variance	Q3	Max	IQR
0	BTC_percent_dif.csv	0.22	26.26	41.40	41.22	1.08	24.973522	623.676820	54.24	103.16	27.97
1	DOGE_percent_dif.csv	0.04	17.36	41.21	37.33	0.04	29.406483	864.741245	61.14	104.02	43.78
2	ETH_percent_dif.csv	0.52	49.55	64.08	66.07	48.92	24.700925	610.135709	80.90	107.35	31.34
3	SOL_percent_dif.csv	0.30	45.14	59.07	59.50	49.73	26.526674	703.664435	77.64	105.27	32.49
4	XRP_percent_dif.csv	0.05	40.20	60.28	63.97	0.77	34.404356	1183.659703	97.22	110.07	57.02

We aimed to test the following hypothesis: **Is there enough evidence to prove that the absolute difference is less than 25%?** If the percent difference was less than 25 percent, there is likely enough evidence to prove that the changes in sentiment on reddit reflect or predict the changes in the cryptocurrency market.

H_0: The mean is greater than 25%. H_A: The mean is less than or equal to 25%.

We will be using a significance level of 0.05. So, if our calculated p-values (using `p_value.py`) are less than .05, we will reject the null hypothesis. Below are calculations for each cryptocurrency's p-value.

In [1]:
```
import p_value

print(f"BTC p-value: {p_value.calculate_p_value(41.21, 29.406483, 186)}")
print(f"DOGE p-value: {p_value.calculate_p_value(41.4, 24.973522, 186)}")
print(f"ETH p-value: {p_value.calculate_p_value(49.55, 24.700925, 186)}")
print(f"SOL p-value: {p_value.calculate_p_value(59.07, 26.526674, 186)}")
print(f"XRP p-value: {p_value.calculate_p_value(60.28, 34.404356, 186)}")
```

```
BTC p-value: 0.9999999999988379
DOGE p-value: 0.9999999999999999
ETH p-value: 1.0
SOL p-value: 1.0
XRP p-value: 1.0
```

As seen above, our p-values are all greater than 0.05. This means that we fail to reject H_0 as there is not enough evidence to support that the mean is less than 25%. Therefore, according to our data, there is **not** a correlation between the sentiment expressed on cryptocurrency subreddits and the value of the cryptocurrency.

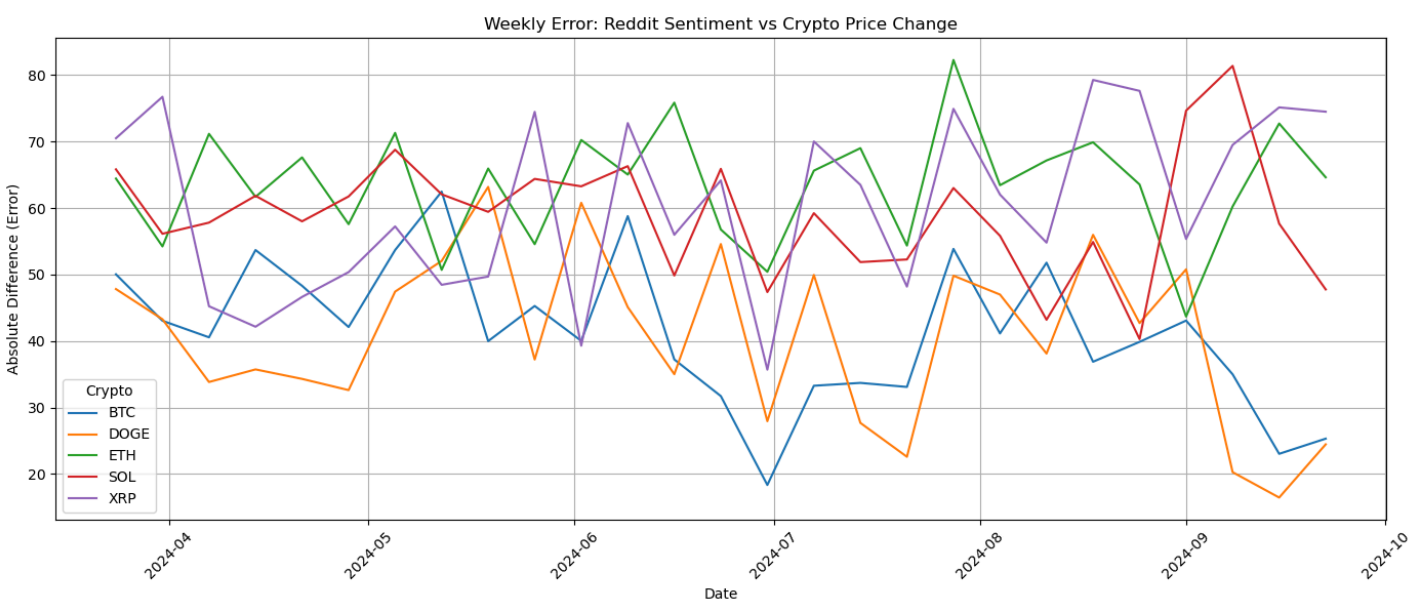
Visualizations

Absolute Error By Week

In [ ]:
```
%matplotlib inline

import statistical_analysis.error_graph as eg
import importlib
importlib.reload(eg)

eg.plot_error_graph()
```

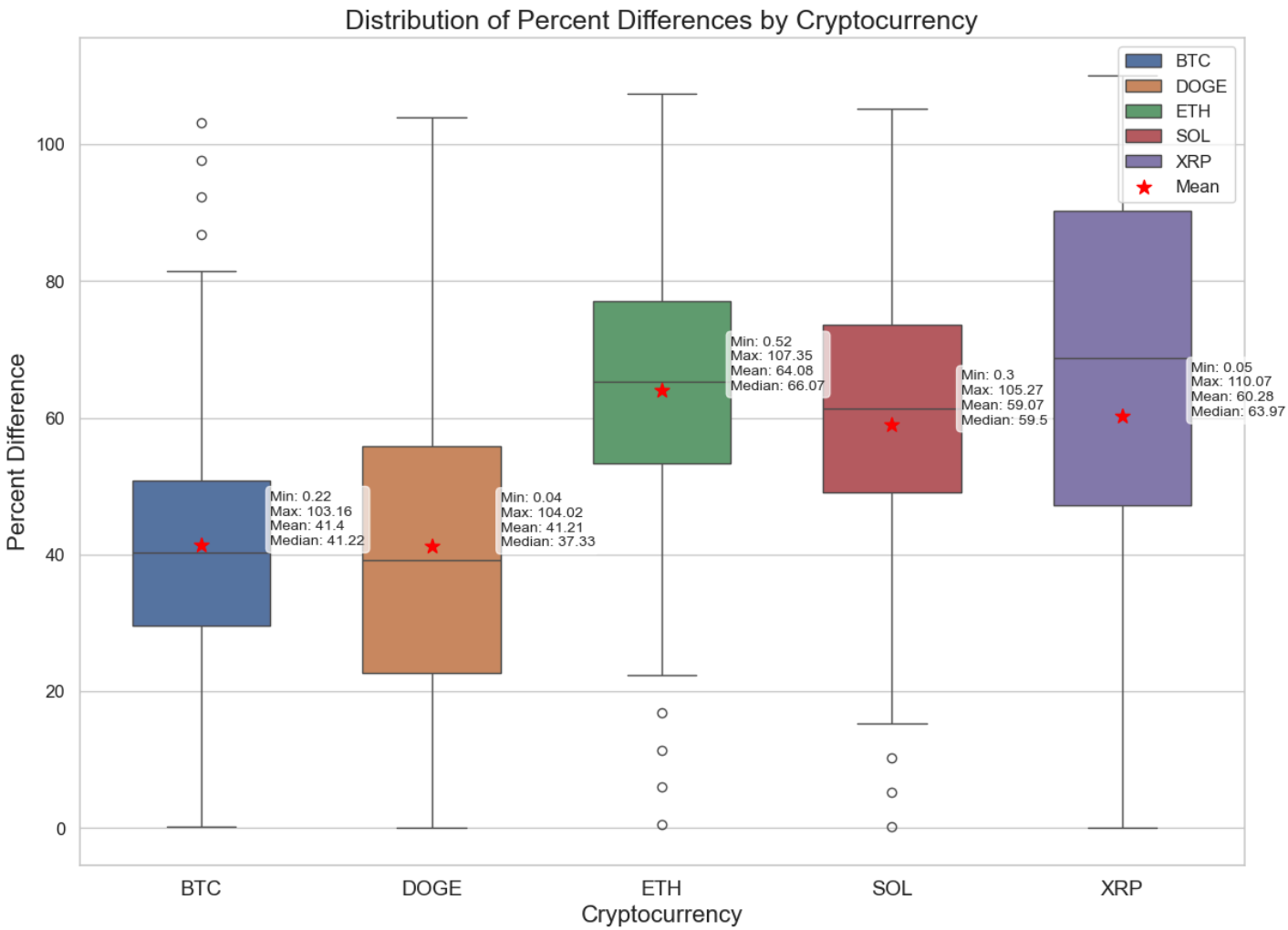


The line graph shows the weekly absolute error between Reddit sentiment and crypto price changes. BTC and DOGE have the smallest median errors, while ETH shows the largest (~82%), with XRP and SOL not far behind. All coins show significantly lower errors around July, suggesting a closer match between sentiment and actual price

changes. DOGE and BTC both trend toward lower error rates in the final weeks, while ETH and XRP have more frequent and larger shifts

Distribution of Percent Differences

In [2]:
```
from modeling.Modeling import create_crypto_boxplot
fig = create_crypto_boxplot()
```




```
<Figure size 640x480 with 0 Axes>
```

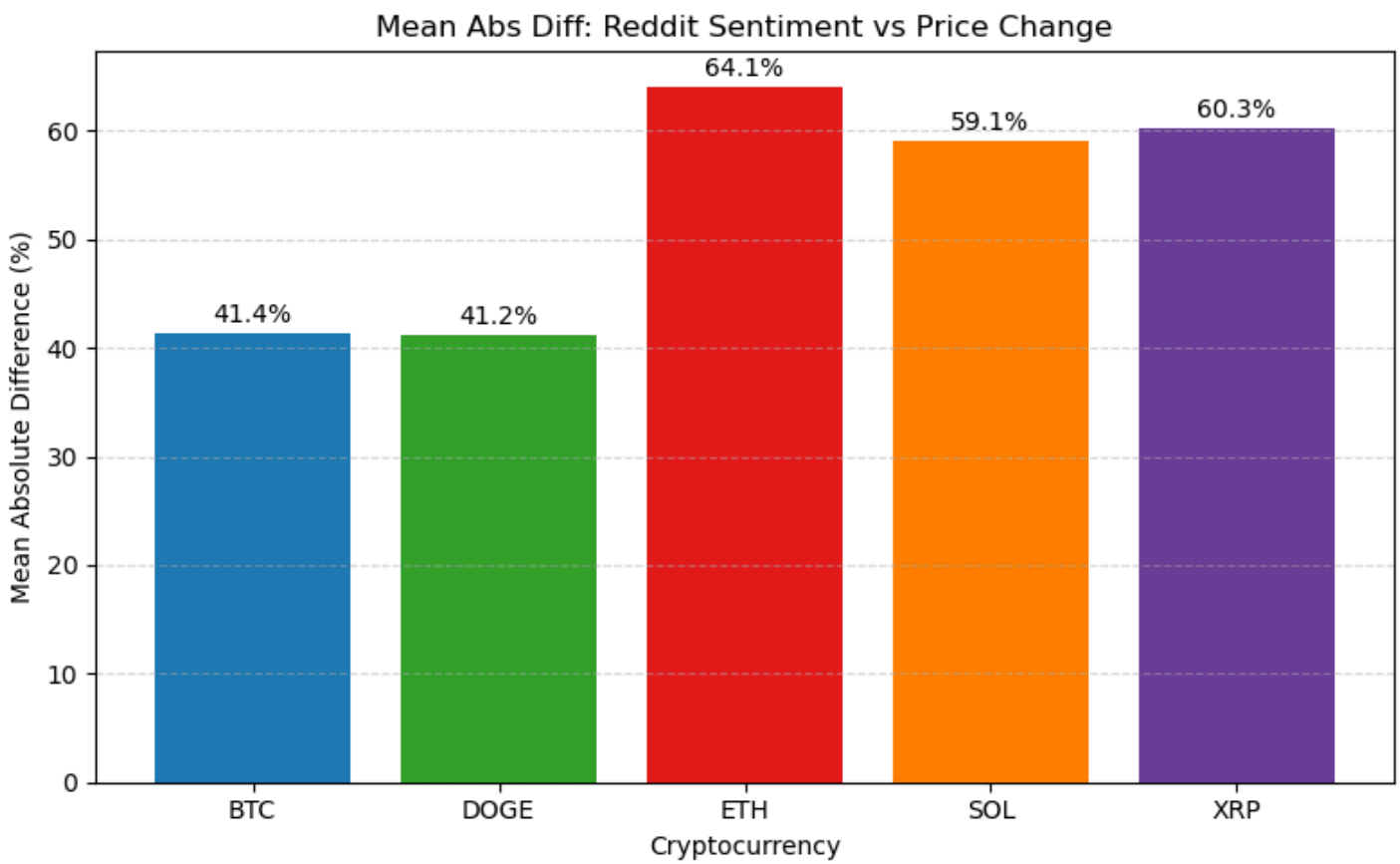
The box plot shows the distribution of percentage differences by Crypto coin. It shows the outliers present for BTC,ETH and SOL along with other key factors like Mean, Median, Min and Max for each Crypto coin individually. Bitcoin and Dogecoin have the smallest median differences between Reddit sentiment and price movements (41.22% and 37.33% respectively). Ethereum shows the highest median difference (66.07%), with XRP (63.97%) and SOL (59.5%). BTC displays several high outliers above 80%, while ETH and SOL show low outliers below 20%. XRP has the widest interquartile range, showing the highest variability in the sentiment-price alignment. The red stars marking the mean shows that for DOGE, the distribution is right-skewed while ETH is slightly left-skewed.

Mean Absolute Difference (Reddit Sentiment vs Price Change)

In [1]:
```
%matplotlib inline

import statistical_analysis.mean_diff_bar as mdf
import importlib
importlib.reload(mdf)
```

```
mdf.plot_mean_diff(
    summary_csv='statistical_analysis/summary_stats.csv',
    colors=['#1f78b4', '#33a02c', '#e31a1c', '#ff7f00', '#6a3d9a']
)
```



This bar chart shows how far off Reddit sentiment is from price moves on average, for each crypto. Bitcoin and Dogecoin have the smallest gaps (about 41 % each), meaning when Reddit gets more positive or negative, their prices tend to follow along more closely. Ethereum has the biggest gap (around 64 %), with Solana (~59 %) and XRP (~60 %) not far behind. In other words, shifts in Reddit sentiment line up best with Bitcoin and Dogecoin prices, but for Ethereum, Solana, and XRP the link is weaker, maybe because those markets move for other reasons besides just what people are talking about online.

Correlation Analysis

See the Pearson Correlation calculation results below.

In [1]:
```
from pearson import calculate_pearson_correlation

calculate_pearson_correlation()
```

```
BTC: Pearson Correlation = 0.0849
ETH: Pearson Correlation = 0.0332
DOGE: Pearson Correlation = 0.0022
SOL: Pearson Correlation = 0.0574
XRP: Pearson Correlation = 0.0149
```

As seen above, all the Pearson Correlations are very small and close to 0. This proves a very weak correlation between % difference in positive posts and % difference in crypto price per day. BTC still shows the highest correlation, but it is still a weak one. This aligns with the rest of our results and analysis, proving little to no correlation.

Results

Our project aimed to explore the relationship between Reddit sentiment and cryptocurrency price changes. Overall, we found that BTC and DOGE had the smallest mean absolute difference (~41%), suggesting a somewhat stronger (but still weak) correlation between Reddit sentiment and price movement compared to ETH, SOL, and XRP, which had mean differences closer to 60%. The boxplot visualization confirmed this, showing lower median percent differences for BTC and DOGE. However, there was high variability across all currencies, with significant outliers. Notably, around July 2024, there was a visible crash in positive sentiment across all cryptocurrencies, corresponding to a general crypto market downturn.

Our hypothesis testing indicated that the mean differences were unlikely to be less than 25%, meaning that we cannot confidently claim a strong correlation between Reddit sentiment and price changes. Even after adjusting our methodology from daily to weekly percent differences to smooth out noise, the results remained inconsistent. The weekly error graph shows that while BTC and DOGE errors were somewhat lower and more stable over time, there was still considerable fluctuation, especially for ETH, SOL, and XRP.

If given more time, an improvement would be to diversify sentiment sources beyond Reddit, for example, incorporating Twitter sentiment or news headline analysis via APIs. This could provide a more comprehensive sentiment dataset and potentially uncover stronger or more nuanced correlations. Additionally, implementing more advanced sentiment analysis models, like those using transformer-based NLP models, could improve the accuracy of sentiment labeling and possibly show a correlation between public sentiment and cryptocurrency values.